

Robot Institute of America

The Robot Institute of America, in 1979, defined a robot as 'A reprogrammable, multifunctional manipulator designed to move material, part, tools, or specialized devices

Indian Robot Olympiad

Three young girls from the Canadian International School in Yelahanka, near Bangalore, won the Indian Robot Olympiad held in September 2010

Feature

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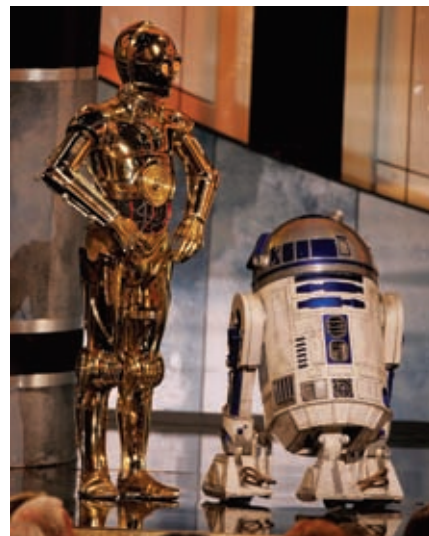
Star Wars fans will remember the scene in *The Empire Strikes Back* when C3PO, reprimanding R2-D2, says, "R2-D2, you know better than to trust a strange computer!" Since Lucas and Spielberg began to make Hollywood movies, scenes like these have ensured that AI and robots have become a part of the popular imagination. It's a-bot time (pardon the shameless pun) that someone separated fact from fiction. So if you've ever wondered where we actually stand in the field of robotics today, read on.

The robot story

Although there may be several ways of defining what a robot is, we like the simple way Robin Murphy, a Professor of computer science and engineering at Texas University, puts it – "a

mechanical creature which can function autonomously". Robots don't have to be anthropomorphic (human-like) or bestial (animal-like), but the bottom line is, they should be able to (a) do things on their own with minimal or no human intervention, and (b) adapt to changes in their environment.

Leonardo Da Vinci is credited with the first design concept of a robot, around 1495. William Grey Walter's Elmer and Elsie (1948) were the first electronic autonomous robots that had touch and light sensors, two vacuum tubes, basic analogue circuits and could even recharge their own batteries. Shakey, the first mobile robot that had a form of AI, was built by the Stanford Research Institute between 1967 and 1970. Shakey could move, turn around, and plan its own route, using bump sensors, a TV camera, a triangulating range finder and was connected to DEC Programmed Data Processor microcomputers (PDP-10



The popular perception of Robots is often based on Hollywood representations such as these

and PDP-15) computers via radio.

Primarily, the four areas where robots have been most useful are manufacturing, space exploration, the

AcYUT. Fateh is dear to us because he's a part of the Digit community, and at just 17 is already building robots all by himself. He started playing with robots when he was still in school, and is proficient with C and micro-controller level programming, and also putting together the hardware.

Fateh credits Thinklabs at IIT Bombay for his foray into robotics. Not only did they help him hone up his skills in programming, they also got him familiar with the building blocks involved. Dr. Prahlad Vadakkepat, General Secretary, FIRA, believes youngsters such as Fateh, "are an



Fateh and his robot



example of the passion that the young population of Indian students display. India has immense talent and potential, however, many of these are unable to tap into their true potential for lack of

resources or opportunities."

FIRA 2010 also marked the foundation of FIRA India, and with it, the path for a national robotics event on a yearly basis. Hopefully, this would serve as the much needed fillip for universities and colleges to come together under a single umbrella to showcase their

innovations and technical prowess in the field of robotics. Janesh Janardhanan, Secretariat and co-host, FIRA 2010 echoes these sentiments and is quite optimistic about the opportunity the

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